

COMP9414 Artificial Intelligence

Additional exercises: reasoning with uncertainty

UNSW Sydney

1 Instructions

These exercises are extra practice for Week 7. They deliberately use different examples from the main tutorial handout, while staying close to the same ideas: Bayes' rule, Bayesian networks, exact inference, decisions under uncertainty, Bayes filters, and calibrated confidence.

Exercises 1–6 and 8 can be completed by hand. Exercise 7 uses a standard Python environment plus scikit-learn.

For Exercise 7, place the supplied Python starter file in your working directory, then install the required dependencies:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install numpy scikit-learn matplotlib
python reasoning_with_uncertainty_calibration_exercise.py
```

On Windows PowerShell, use:

```
py -3.11 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install numpy scikit-learn matplotlib
python reasoning_with_uncertainty_calibration_exercise.py
```

Working convention. Show the probability expression you are evaluating before substituting numbers. For Bayes-filter questions, keep at least three decimal places until the final normalisation.

2 Exercise 1: Design a robot safety Bayes net

A mobile robot is deciding whether a corridor segment is risky. Use these Boolean variables:

Variable	Meaning
R	it rained heavily before the delivery run
W	the floor near the entrance is wet
C	the corridor is crowded
B	the camera image is blurred
S	the wheels slip during the first metre
H	the segment is hazardous for autonomous travel

Assume the following direct influences:

- heavy rain can make the floor wet and can blur the camera image;
 - crowding can blur the camera image and can make the segment hazardous;
 - a wet floor can cause wheel slip and can make the segment hazardous;
 - wheel slip can make the segment hazardous.
1. Draw a Bayesian network that matches these assumptions.
 2. Write the factorisation of the full joint distribution $P(R, W, C, B, S, H)$ implied by your graph.
 3. Which variables need prior probability tables?
 4. Which variables need conditional probability tables with two parents?
 5. Give one conditional independence claim that follows from your graph.
 6. Give one pair of variables that may become dependent after conditioning on H . Explain the “explaining away” intuition in one or two sentences.
 7. Suppose the camera image is blurred. Does that observation prove the floor is wet? Explain the difference between evidence and certainty.

3 Exercise 2: Exact inference by enumeration

Consider a small diagnostic Bayes net for a delivery robot:

$$B \rightarrow S \leftarrow M, \quad M \rightarrow D.$$

The variables are:

B = battery low, M = motor fault, S = robot moves slowly, D = diagnostic warning.

The priors are $P(B) = 0.25$ and $P(M) = 0.10$. Assume B and M are independent before any evidence is observed.

B	M	$P(S \mid B, M)$		M	$P(D \mid M)$
true	true	0.95		true	0.90
true	false	0.80		false	0.20
false	true	0.70			
false	false	0.05			

1. Write the factorisation of $P(B, M, S, D)$.
2. Compute $P(S)$ by summing over all assignments of B and M .
3. Compute $P(M \mid D)$. Which term in the Bayes rule expression is the normalising constant?
4. Compute $P(B \mid S)$ by enumeration.
5. Compute $P(B \mid S, D)$. Compare this with $P(B \mid S)$ and explain the direction of the change.
6. Is B independent of D before observing S ?
7. Is B independent of D after observing S ? Explain using the collider structure $B \rightarrow S \leftarrow M$.

4 Exercise 3: Expected utility and value of information

A robot can send a fragile sample through a hallway. Let G mean the hallway is genuinely clear. The robot's current calibrated belief is $P(G) = 0.65$. It has three immediate actions:

Action	Utility if G true	Utility if G false
Go now	80	-120
Wait for a human helper	20	20
Use a longer route	35	35

1. Compute the expected utility of each immediate action.
2. Which action should a risk-neutral agent choose?
3. For what values of $p = P(G)$ does **Go now** beat **Use a longer route**?
4. For what values of $p = P(G)$ does **Go now** beat **Wait for a human helper**?
5. A supervisor can inspect the hallway and reveal G perfectly before the robot acts. Ignoring inspection cost, compute the expected utility with perfect information.
6. Compute the value of perfect information.
7. If asking the supervisor costs 12 utility points, should the robot ask?
8. Explain why the calculation depends on calibrated probabilities, not just confident probabilities.

5 Exercise 4: A different Bayes filter

This exercise uses a six-cell corridor rather than the five-cell door example from the tutorial. The robot's hidden state X_t is its cell:

$$X_t \in \{1, 2, 3, 4, 5, 6\}.$$

Cells 2 and 5 contain a charging beacon. The prior belief is:

$$\text{bel}_0 = [0.05, 0.10, 0.25, 0.30, 0.20, 0.10].$$

The binary sensor reports whether it sees a beacon:

$$P(z = \text{beacon} \mid \text{beacon cell}) = 0.75, \quad P(z = \text{beacon} \mid \text{non-beacon cell}) = 0.15.$$

The robot then commands **move left**. Its motion model is:

Outcome	Probability
moves one cell left	0.70
stays in the same cell	0.20
overshoots two cells left	0.10

If the move would pass cell 1, clamp the result to cell 1.

1. Perform the sensor update for $z_1 = \text{beacon}$. Write the unnormalised vector and the normalised posterior.

2. Apply the `move left` prediction step to your posterior.
3. Perform another sensor update for $z_2 = \text{no beacon}$.
4. Which cell is most likely after these two observations and one action?
5. Why should the robot still keep a full belief distribution rather than only the most likely cell?
6. If a programmer wanted to simulate this filter, what data structures would they need for the belief state, sensor model, and motion model?

6 Exercise 5: Sensor reliability in a Bayes filter

This exercise stays with the Bayes-filter idea from the lecture, but asks how the belief changes when the sensor model is more or less reliable.

A robot is in one of four numbered zones:

$$X_t \in \{1, 2, 3, 4\}.$$

Zones 2 and 4 contain reflective signs. The prior belief is:

$$\text{bel}_0 = [0.40, 0.25, 0.25, 0.10].$$

The robot observes $z_1 = \text{reflection}$. Two possible sensor models are:

Sensor model	$P(\text{reflection} \mid \text{sign})$	$P(\text{reflection} \mid \text{no sign})$
Trusted sensor	0.85	0.10
Degraded sensor	0.60	0.35

1. Compute the posterior belief after $z_1 = \text{reflection}$ using the trusted sensor model.
2. Compute the posterior belief after the same observation using the degraded sensor model.
3. Which posterior is more concentrated on the reflective-sign zones? Why?

Now continue from your trusted-sensor posterior. The robot commands `move right`. Its motion model is:

Outcome	Probability
moves one zone right	0.70
stays in the same zone	0.20
slips one zone left	0.10

Clamp moves at zones 1 and 4.

4. Apply the motion update to the trusted-sensor posterior.
5. The next observation is $z_2 = \text{no reflection}$. Apply the trusted sensor model again and compute the final posterior.
6. Which zone is most likely after the two observations and one action?
7. Suppose the sensor was actually degraded, but the robot kept using the trusted sensor model. Describe one way this could make the robot's behaviour overconfident or unsafe.

7 Exercise 6: Calibration by hand

A classifier predicts whether a robot delivery will require human intervention. The table gives the model's predicted probability for the positive class and whether the positive class actually occurred.

Case	Predicted p	Outcome	Case	Predicted p	Outcome
1	0.52	1	9	0.85	1
2	0.58	0	10	0.88	0
3	0.62	1	11	0.91	1
4	0.66	0	12	0.93	1
5	0.71	1	13	0.95	0
6	0.74	1	14	0.97	1
7	0.78	0	15	0.99	1
8	0.82	1			

Use the bins $[0.5, 0.7)$, $[0.7, 0.9)$, and $[0.9, 1.0]$. For each bin, compute:

average confidence, fraction positive, $|\text{fraction positive} - \text{average confidence}|$.

1. Complete the calibration table for the three bins.
2. For each bin, say whether the model is overconfident, underconfident, or roughly calibrated.
3. Compute the expected calibration error:

$$\text{ECE} = \sum_b \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|.$$

Here $\text{acc}(B_b)$ means fraction positive in the bin.

4. Why can a model have high classification accuracy but poor calibration?
5. Give one decision in an AI system where poor calibration would matter even if the top-1 prediction is often correct.

8 Exercise 7: scikit-learn calibration lab

Open the starter file:

`reasoning_with_uncertainty_calibration_exercise.py`

Use the setup commands from the instructions section, then run:

`python reasoning_with_uncertainty_calibration_exercise.py`

The script trains a logistic regression model, an uncalibrated random forest, and calibrated random forests on a synthetic imbalanced classification problem. It prints accuracy, Brier score, log loss, and expected calibration error.

Metric reminder. **Accuracy** measures how often the final class prediction is correct. **Brier score** is the mean squared error of the predicted probability for the true binary outcome,

$$\frac{1}{n} \sum_i (p_i - y_i)^2,$$

so lower is better. **Log loss** is

$$-\frac{1}{n} \sum_i (y_i \log p_i + (1 - y_i) \log(1 - p_i)),$$

so it heavily penalises confident wrong probabilities and lower is better. **Expected calibration error** compares average confidence with observed frequency across probability bins, so lower means the probabilities are better calibrated by that binning.

1. Record the metrics for each model.
2. Which model has the best Brier score? Is it also the most accurate?
3. Which model has the best log loss? Does this match the Brier-score ranking?
4. Compare the uncalibrated and calibrated random forests. What changed?
5. Change `CLASS_SEP` in the script from 0.65 to 0.35. How do the metrics change when the classes overlap more?
6. Change `N_BINS` from 10 to 5. Why can the measured expected calibration error change even when the model predictions are the same?
7. Inspect the saved reliability plot. Which curve is closest to the diagonal?

9 Exercise 8: Abstention from calibrated confidence

An answer-selection system estimates $q = P(\text{answer is correct})$. If it answers and is correct, utility is +2. If it answers and is wrong, utility is -8. If it abstains, utility is 0.

1. Derive the expected utility of answering as a function of q .
2. What is the minimum q for which answering has positive expected utility?
3. Apply your threshold to these calibrated confidence estimates:

0.54, 0.63, 0.71, 0.79, 0.83, 0.88, 0.94.

Which cases should the system answer?

4. Suppose these q values are not calibrated probabilities but raw neural-network scores. What part of your decision rule becomes unsafe?
5. How could Exercise 7 help you build a more defensible abstention policy?

10 Submission checklist

Your submission should include:

- the Bayes-net graph and factorisation from Exercise 1;
- probability calculations for Exercises 2, 4, 5, and 6;
- expected-utility and value-of-information calculations for Exercise 3;
- short written explanations for the calibration and abstention questions;
- scikit-learn metrics and reliability plot observations from Exercise 7.